Faculty of Science, Engineering and Built Environment

SEE332 Transmission and Distribution System Design

Deakin University Unit Guide

Trimester 1, 2025

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Welcome

This is a project oriented designed based learning (PODBL) unit, which aims to provide students the opportunity to gain a solid understanding of the operating principles and design of power transmission and distribution systems. Students will engage in a Transmission and Distribution System Design Project where they will design active and reactive power controllers as well as voltage compensation devices to mitigate the impact of renewable energy sources on the transmission and distribution. They will explore and learn about different transmission and distribution system technologies. Students will develop the concepts and skills on transmission and distribution line modelling, control mechanisms to improve their efficiency as well as the impact of renewable energy sources and new types of loads on the transmission and distribution network. Students will also develop skills in computer modelling and practical implementation of their design to analyse the transmission and distribution system performances. The hands-on skills developed through this unit will help students implement advanced technologies for ensuring effective operation of the electricity transmission and distribution networks in the presence of renewable energy sources.
This Unit Guide provides you with the key information about this unit. Please read it carefully and refer to it frequently throughout the study period. Your unit site also provides information about your rights and responsibilities. We will assume you have read this before the unit commences, and we expect you to refer to it throughout the study period.

To be successful in this unit, you must:

- read all materials in preparation for your learning activities and follow up each with further study and research on the topic
- start your assessment tasks well ahead of the due date
- read or listen to all feedback carefully and use it in your future work
- attend and engage in all educator facilitated (scheduled) learning activities and other learning experiences as part of the unit design

Who is the unit team?

Unit chair: leads the teaching team and is responsible for overall delivery of this unit

Name: Enamul Haque
Campus: Geelong Waurn Ponds,
Email: enamul.haque@deakin.edu.au
Phone: +61 3 522 71206
Other members of the team and how to contact them

Name: Mr James Lamont, Technical Officer, (Electrical Engineering) for Lab support
Campus: Waurn Ponds, Geelong
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Phone: +61 3 52479450

Administrative queries

- check-out the 'SEBE Student Hub' section on your unit site
- contact your Unit Chair or Campus Leader
- drop in or contact Student Central to speak with a Student Adviser

For additional support information, please see the Rights and Responsibilities section under 'Content' in your unit site.
About this unit

Unit development in response to student feedback

Every trimester, we ask students to tell us, through eVALUate, what helped and hindered their learning in each Unit. You are strongly encouraged to provide constructive feedback for this Unit when eVALUate opens (you will be emailed a link).

In previous versions of this unit, students have told us that these aspects of the Unit have helped them to achieve the learning outcomes:

- Practical lab work helped them to learn more effectively about practical transmission and distribution system.
- Detailed lecture notes were very useful in understanding content.
- Project work helped learn transmission and distribution system technologies used in power system.

The following aspects of the unit have been introduced, enhanced or retained in response to feedback from students who have undertaken this unit in previous trimesters:

- Detailed Matlab modelling of individual components for the project work during studio sessions, to provide more help to students.
- Additional help for Matlab/Simscape for transmission and distribution system application.

If you have any concerns about the Unit during the trimester, please contact the unit teaching team - preferably early in the trimester - so we can discuss your concerns, and make adjustments, if appropriate.

Learning Outcomes

Each Unit in your course is a building block towards Deakin's Graduate Learning Outcomes - not all Units develop and assess every Graduate Learning Outcome (GLO).

ULO	These are the Unit Learning Outcomes (ULOs) for this unit. At the completion of this unit, successful students can:	Alignment to Deakin Graduate Learning Outcomes (GLOs)
ULO1	Apply technical knowledge and demonstrate competency to design transmission and distribution systems.	GLO1: Discipline-specific knowledge and capabilities GLO4: Critical thinking GLO5: Problem solving
ULO2	Validate a transmission and distribution system design through computer modelling software and practical implementation in a laboratory environment.	GLO2: Communication GLO3: Digital literacy
ULO3	Propose solutions to design problems for a transmission and distribution system and communicate the findings through a professional technical report.	GLO1: Discipline-specific knowledge and capabilities GLO2: Communication GLO5: Problem solving
ULO4	Collaborate with a team to develop, implement and complete a project management strategy and reflect upon the development of teamwork skills.	GLO2: Communication GLO7: Teamwork
ULO5	Engage ethically, practice professional conduct and Occupational Health and Safety (OHS) and demonstrate the understanding of safe work practices in a team environment with students from diverse backgrounds.	GLO6: Self-management GLO8: Global citizenship

Assessing your achievement of the unit learning outcomes

Hurdle requirements

To be eligible to pass this unit students must achieve a mark of at least 40% in the project validation demonstration task (AT3), and a mark of at least 50% in the combined Problem solving skill and knowledge tests (AT2)

Brief summary of the hurdle requirement/s	Rationale

To be eligible to pass in this unit, student must • achieve a mark of at least 40% in the project validation demonstration task (AT3) • achieve a mark of at least 50% in the combined Problem solving skill and knowledge tests (AT2) This unit-specific activity may be undertaken either at scheduled session during the trimester (for on campus students) or during the unit-specific activities scheduled during the 'intensive' week (for online students).	The students are required to fulfil the hurdle requirements to achieve the unit learning outcomes (ULOs) and course learning outcomes (CLOs)

Summative assessment (tasks that will be graded or marked)
NOTE: It is *your responsibility* to keep a backup copy of every assignment and the materials used to develop/complete it where possible (e.g. written/digital reports, essays, videos, images). In the unusual event that one of your submissions becomes corrupted, is incorrectly submitted or otherwise lost, you may be asked to submit the backup copy. Any work you submit may be checked by electronic or other means for the purposes of detecting breaches of academic integrity such as collusion, plagiarism and contract cheating. You must understand your responsibility to act with honesty and integrity in your studies as Deakin takes all breaches very seriously. Make sure you read Your rights and responsibilities as a student in this unit to find out more about academic integrity.

Deakin has a universal assessment submission time of 8 pm AEDT/AEST. A late penalty will apply to assessments submitted after 11.59 pm AEDT/AEST.
Summative assessment task 1

Title	**Project plan (team)**
Brief description of assessment task	In this assessment task, students are required to develop a project plan. Students are required to demonstrate a clear understanding of the project work on transmission and distribution network design to be completed during the trimester.
Detail of student output	This is a team assessment task and each team will be required to submit a single project plan (15 pages maximum) which includes: • Background of the project (including a literature review) • Significance of the project • Aims/objectives of the project • Outcome/deliverables of the project • Methodologies • Occupational Health and Safety (OHS) considerations in conducting the practical work, • Team formation and responsibilities • Plan for communication and team management/agreement. • Individual learning goal to develop team work skills • Required resources (including computer modelling & practical activities)
Grading and weighting (% total mark for unit)	10% (Marked)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Apply technical knowledge and demonstrate competency to design transmission and distribution systems. ULO4 - Collaborate with a team to develop, implement and complete a project management strategy and reflect upon the development of teamwork skills
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – by demonstrating the ability to understand the key concepts of power transmission and distribution system design. GLO2 – by demonstrating the ability to communicate with team members and prepare a project plan GLO7- by demonstrating the ability to collaborate with members of a team to develop a detailed project plan.
How and when you will receive feedback on your work	Students will receive feedback on their design, from their facilitator during the weekly, facilitated project team meetings. The project meeting will be held during the weekly seminars for campus students or via Zoom for online students.

When and how to submit your work	**Week 4** - Thursday, 27 March 2025 by 8pm (AEDT) via the unit site.

Summative assessment task 2

Title	**Problem solving skill and knowledge tests (× 2)**
Brief description of assessment task	In this assessment task, students are required to solve a series of design problems and provide answers to conceptual questions individually on transmission and distribution system. The tests will require both written responses and mathematical calculations. Students will apply their understanding of the content covered in Transmission and Distribution System Design, including modelling of transmission and distribution lines, reactive power and voltage control.
Detail of student output	This is an individual task. Students will be required to submit written responses to design problems and conceptual questions. Students are expected to demonstrate their problem solving skills, conceptual understanding and the ability to reflect critically on their designed solutions. Test paper will be available in the unit site. Both on-campus and online students will upload their answers to the Unit site.
Grading and weighting (% total mark for unit)	30% (2 x 15% each) - Marked
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1 - Apply technical knowledge and demonstrate competency to design transmission and distribution systems.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – by demonstrating the ability to apply technical knowledge to provide solution to design problems and answers to the conceptual questions related to the transmission and distribution system. GLO4 – by demonstrating the ability to critically analyse power transmission and distribution system components and their applications. GLO5 – by demonstrating the ability to provide solutions to a given problem
How and when you will receive feedback on your work	Students will receive feedback during problem solving seminars. Students will work through design problems. Students are encouraged to go through lecture notes and tutorial problems discussed during the seminar.
When and how to submit your work	**On-campus and Online students:** **Test 1**: via unit site **Date**: Week 6, Wednesday April 9 2025 **Time:** 6pm to 7.30pm (AEST) **Time Allowed:** 1.5 hours plus 30 minutes for scanning and uploading the test paper **Test 2:** via unit site **Date**: Week 11, Wednesday May 21, 2025 **Time**: 6pm to 7.30pm (AEST) **Time Allowed:** 1.5 hours plus 30 minutes for scanning and uploading the test paper

Summative assessment task 3

Title	**Project validation demonstration**
Brief description of assessment task	In this assessment task, students are required to demonstrate their ability to develop the computer model and validate part of the project practically in the laboratory. Students are required to demonstrate their application of OHS understanding, orally explain their project along with an explanation for the success or failure of their design to solve the provided problem scenario. Students will be able to use the feedback during conversation for further consideration in their final report.

Detail of student output	This is an individual oral and practical task. Each student is required to demonstrate and verbally defend the results during project validation. Students will be assessed on • The development of their computer model and demonstration of the practical system for the part of project. • Their understanding of OHS considerations through verbal communication and the correct use of equipment during the laboratory session
Grading and weighting (% total mark for unit)	25% (Marked)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO1: Apply technical knowledge and demonstrate competency to design transmission and distribution systems. ULO2: Validate a transmission and distribution system design through computer modelling software and practical implementation in a laboratory environment. ULO5: Practice professional conduct and Occupational Health and Safety (OHS) and demonstrate the understanding of safe work practices in an engineering environments.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – by demonstrating the ability to apply their technical knowledge the develop computer model and practical system for the project. GLO2 – by demonstrating the ability to orally communicate the designed project outcomes to a professional audience. GLO3 - By demonstrating the ability to use software technology and collect resources to complete the project GLO4 – by demonstrating the ability to critically analyse power transmission and distribution system components and explain why a system worked or failed. GLO6 – by demonstrating the ability to learn and work independently and taking responsibility to work within OHS requirements. GLO8- by demonstrating the ability to engage ethically and productively with peers from diverse communities and cultures.
How and when you will receive feedback on your work	Students will receive feedback during project validation.
When and how to submit your work	**Week 10:** further details will be provided on the unit site at the start of the trimester.

Summative assessment task 4

Title	**Team Final Report**
Brief description of assessment task	In this assessment task, students are required to clearly and professionally present their project outcome in written technical report which include final design detail, computer model, simulation results and practical results. Detailed analysis of the results need to be included in the final report. Students are required to evaluate their design strengths and weaknesses and reflects on where improvements could be made. While students will work together to develop their final project report, they will also be required to demonstrate their individual performance and clarify their contributions, This will be used to individualise the mark for each team member.

Detail of student output	This is a team based assessment task, however, there are some individual components as outlined below. Each team member will submit an individual final report. The report (40 page maximum) will include: • Background, Project objectives and outcomes • Detailed literature review and theoretical background **(individual)** • Design and theoretical analysis • Computer simulation models • Discussions of simulation results **(Individual)** • Results of practical project validation • Discussions of practical results **(individual)** • Learning experiences, suggested recommendations for improvement, Reflection on their teamwork skills
Grading and weighting (% total mark for unit)	35% (Graded) (26.25% group, 8.75% individual)
This task assesses your achievement of these Unit Learning Outcome(s)	ULO3 - Propose solution to design problems for a transmission and distribution system and communicate the findings through a professional technical report.
This task assesses your achievement of these Graduate Learning Outcome(s)	GLO1 – by demonstrating the ability to understand the key concepts of power transmission and distribution system design. GLO2 – by demonstrating the ability to communicate design solutions to an engineering problem through a written professional technical report
How and when you will receive feedback on your work	Students will normally be provided with feedback within 15 working days of the due date. Students are encouraged to seek feedback from their unit chair if required.
When and how to submit your work	**Week 11: Sunday, 25 May 2025 by 8pm (AEST)** via the unit site.

Formative assessment

(for practice and feedback rather than marks and grades)
In addition to the above, you will have a better chance of success in this Unit if you complete these tasks – you will get valuable experience and feedback.
These activities include:

- Team meeting
- Consultation with the lecturer
- Quizzes on unit content

Your learning experiences in this unit

Educator-facilitated (scheduled) learning activities - on-campus unit enrolment

2 x 2 hour seminars per week, 1 x 2 hour practical experience (studio) per week.
Educator-facilitated (scheduled) learning activities - online unit enrolment

Online independent and collaborative learning including a 1 x 2 hour seminar per week. Students are required to attend and participate in practical experiences (laboratories) at the Geelong Waurn Ponds Campus for the scheduled sessions during the trimester intensive activities as detailed in the unit site.
Typical study commitment

Students will on average spend 300 hours over the teaching period undertaking the teaching, learning and assessment activities for this unit.
This will include educator guided online learning activities within the unit site.
Learning and assessment activities for campus students, this include seminars, studios, designated activities in the studio

sessions, assessment tasks, problem-based exercises and project team meetings. Students will also be required to engage weekly with the pre-recorded content and reading materials available in the unit site. The pre-recorded content will support students learning during the discussion and Q & A sessions in the weekly seminars. Students should follow the weekly checklists available on the unit site.

For online students, this time should be divided among engagement with online lecture materials, online learning activities, discussion boards, assessments tasks, readings, study time, project team meetings and practical sessions during the intensive week.

Note

At Deakin, courses are delivered within a learning environment that provides all students with equitable and consistent access to facilities, infrastructure, resources, and support to assist student progress and achievement of learning outcomes.

We have introduced new terms to reflect learning activities to enhance your learning experience, aligning with our innovative **DeakinDesign learning principles and practices**. The new terms better reflect how teaching teams will guide you through your learning journey and the types of learning experiences you will have.

'Lectures' are the activities where teaching staff engage you through presentations with student participation.

In **'seminars'**, an educator will guide you in a smaller group of students through highly interactive discussions and activities.

Your units may also include '**practical experiences**' such as '**laboratory**', '**workshops**', '**clinical skills**' and more. These hands-on activities typically take place in specialised facilities with industry tools, equipment or technology to allow you to apply your knowledge practically.

Some other terms

If you see a '**meeting**' in your timetable, this is an optional drop-in session.

'Assessments' or **'team-based learning**' indicate an in-class evaluation of your skills or knowledge. A **'pre-assessment practice'** could be scheduled to prepare you for these assessments.

Find out more

Take a look at the **Learning activities webpage** for a full list of the terminology changes and reasons they were changed.

Unit learning resources

Your unit learning resources can be accessed from your unit site.
The texts and reading list for SEE332 can be found via the University Library.

Note: Select the relevant trimester reading list. Please note that a future teaching period's reading list may not be available until a month prior to the start of that teaching period so you may wish to use the relevant trimester's prior year reading list as a guide only.

Essential learning resources

There is no prescribed textbook. The following resources will be provided to students:

- Lecture notes, design problems/tutorials, project details, laboratory instructions which are available in the unit site
- Technical/academic facilitator, laboratory facilities, library.

Recommended learning resources

The Deakin Software Library provides students with access to software that you may need or find useful for your study at Deakin.

Recommended reference books

1. Hadi Saadat, “Power System Analysis”, PSA Publication, 3rd Edition, 2010.
2. Turan Gornen "Electrical Power Transmisison System Engineering- Analysis and Design" Third Edition, CRC Press, 2014
3. Turan Gornen "Electrical Power Distribution Engineering" Third Edition, CRC Press, 2014.
4. D. Glover, T. J Overbye and M. S. Sarma “Power System Analysis and Design”, 6th Edition, Cenage Learning.
5. V. K. Mehta “Principle of Power System”, 3rd Edition, S. Chand, March, 2005
6. D.W Hart "Power Electronics", McGrawhill, 2010.

Recommended Links

1. https://www.epri.com Electric Power Research Institute
2. https://www.theiet.org The Institution of Engineering and Technology
3. Institute of Electrical Electronic Engineers, USA, Link: www.ieee.org

Access IEEE Xplore through Deakin website to collect journal papers, conference papers and IEEE standards.

Where to access unit resources

Textbooks can be sourced from various outlets including direct from the publisher, online bookshops, or retailers. Limited copies of textbooks are also available on loan from the University Library.

Key dates for this study period

Trimester 1 teaching period begins	Monday 3 March 2025
Census date	Monday 31 March 2025
Easter/intra-trimester break	Friday 18 April - Sunday 27 April 2025
Trimester 1 teaching period ends	Friday 23 May 2025
Study period (end-of-unit assessment /examination preparation period)	Monday 26 May - Friday 30 May 2025
End-of-unit assessment and examination period	Monday 2 June - Friday 13 June 2025
Inter-trimester break (the period between trimesters)	Monday 16 June - Friday 4 July 2025
Unit results released	Thursday 3 July 2025 (10.30 am)

Unit weekly activities

Week	Commencing	Topic/Task	Learning activities	Assessment due date
1	3 March 2025	• Unit outline • Introduction to Power System • Review of AC power analysis • Synchronous Generator	Lab induction Seminar studio	
2#	10 March 2025	• Transmission line components • Transmission line parameters • Design considerations for transmission line • Project work	Seminar studio	
3	17 March 2025	• Transmission line modelling -Part I • Transmission line modelling -Part II • Project work	Seminar studio	
4	24 March 2025	• Complex Power flow through Transmission line • Reactive Power and Voltage Control Methods • Project work	Seminar studio	**Assessment 1: Project plan (team)** due Thursday 27 March

5	31 March 2025	• FACTs Devices • Project work	Seminar studio	
6	7 April 2025	• HVDC Transmission system • Project work • Problem Solving Skill and knowledge Test 1	Seminar studio	**Assessment 2: Problem solving skill and knowledge test 1** Wednesday 9 April
7*~	14 April 2025	• Introduction to Power Distribution System • Design Considerations for distribution system • Project work	Seminar studio	
8	28 April 2025	• Applications of Transformers in distribution system • Project work	Seminar studio	
9	5 May 2025	• Applications of Capacitor in Distribution System & Voltage regulation • Project work	Seminar studio	
10	12 May 2025	• Impacts of Renewable Energy Sources and Electric Vehicle on Power Grid • Project work • Problem Solving Skill and Knowledge Test 2	Seminar studio	**Assessment 3: Project validation presentation** due date details provided on unit site
11	19 May 2025	• Project work • Final Project Report	Seminar studio	**Assessment 2: Problem solving skill and knowledge test 2** Wednesday 21 May **Assessment 4: Team Final Report** due Sunday 25 May
12	26 May 2025			Study period
13-14^	2 June 2025			End-of-unit assessment and exam period

Labour Day public holiday: **Monday 10 March 2025** - University closed
* Easter/Intra-trimester break: **Friday 18 April - Sunday 27 April 2025** (between weeks 7 and 8)
~ ANZAC Day public holiday: **Friday 25 April 2025** - University closed
^ King's Birthday public holiday: **Monday 9 June 2025** - University closed